

KitchenSink Evaluation

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Qualitative Evaluation

For the qualitative evaluation, we wanted to understand how real users experience our system on a functional level, but also whether it felt intuitive or genuinely useful for everyday life. We reached out to the initial five user participants that were familiar with our system as well as NPPR and Crum Ledge residents on Swarthmore College's campus and off-campus residents because they maintain their own fridges with actual ingredients, which represents more authentic use case for a fridge-to-recipe application. Participants were recruited via email, which introduced the system and invited them to use it followed by a Google Form.

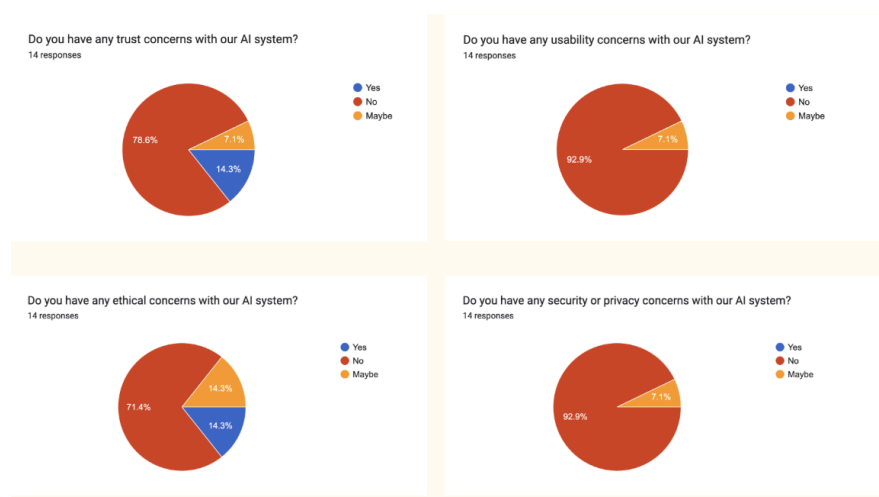
The form asked participants to reflect on their experience using a Likert scale on whether the app was easy to use, if they would integrate it into their daily routine, how appetizing the recipes seemed. We asked how difficult they found each stage of the process with a yes, no, a little for onboarding, uploading photos, editing ingredients, generating recipes, viewing recipes, and regenerating recipes. We also asked what specifically worked in the application offering a structured checklist that included self-explanatory process, user-friendly interface, comprehensive onboarding, quality of ingredients detected, and the utility of the fridge reference view during ingredient editing. Lastly, we asked four yes or no questions regarding whether users had trust, ethical, usability, or security and privacy concerns about our AI system. We reviewed the responses across 14 participants and identified emerging themes. We looked for both positive and negative/constructive feedback.

Based on the responses, we concluded that the system is successful to its core purpose and feels accessible to new users (92.9%), while also revealing some limitations in

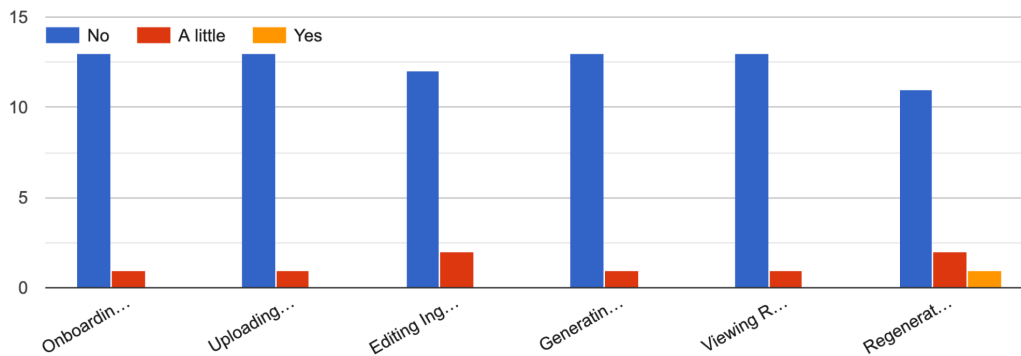
personalization that would like to address before the system could be deployed for the general public to use .

The most consistent positive theme was that the system was self-explanatory. From the feedback, 100% of users felt that they could navigate through our application without needing guidance and that it was intuitive to use. However, feedback on the interface itself was more divided. While users could move through the application, several noted that the UI felt that it could be improved. This is something that we should consider as we look at the future development of our project. If a system feels unpolished to users, they might not want to return to using it again. This becomes a friction point that generally erodes the motivation to engage with the application.

The most important finding came through the ethical and trust concern responses which showed more variability than the usability questions. While 71.4% of users did not have ethical concerns, 14.3% did and 14.3% might have ethical concerns. We had a user that was vegetarian receiving meat recipes because the shared fridge contained meat belonging to roommates. From a user's perspective, the system ignored their dietary preferences which undermines the user's autonomy and creates a sense of disconnection for the user and the application.



Did you find it difficult to navigate the following:



Quantitative Evaluation:

For our quantitative evaluation, we wanted to see how well the model was at identifying ingredients from the fridge pictures. To do this, we first created ground-truth labels by manually labeling five different fridge pictures. The pictures ranged in organization, where some were extremely filled and cluttered while others were more sparse. We then provided the model with each picture to create the list of ingredients identified by the model.

After having our model-generated and manually labeled lists, we determined accuracy using Jaccard similarity and LLM-as-a-Judge as metrics. Jaccard similarity was used to compare how similar two sets are. However, due to concern about differences in wording such as “water” and “water bottle” or “eggs” and “egg,” we chose to use LLM-as-a-Judge to avoid issues with semantic differences. In using the LLM as a judge, we prompted the model to ignore minor differences when calculating accuracy.

Each of the five images were run through the VLM five times since responses could differ slightly from each run and accuracy was calculated for each image and then aggregated as average accuracy for each run. Using Jaccard similarity as our metric, the model had an average accuracy of 0.278. This means that our model was only able to identify 27% of the ingredients in

the picture. However, with LLM-as-a-Judge, our ingredient detection had an average accuracy of 0.592. This means that around 59% of the ingredients were able to be detected. The gap between Jaccard and LLM-as-a-judge suggests that while the model can identify the ingredients, it may not exactly match the ground-truth labels, highlighting how Jaccard may be too strict of a metric to evaluate our model's performance at identifying ingredients.

